

## SRS Rough Draft

### Functional requirements

#### FR-01 | File selection

The system shall allow the user to select a BasicML text file from any accessible location through a graphical file picker.

#### FR-02 | Program loading

The system shall load the words of a valid BasicML file into consecutive memory locations beginning at address 00.

#### FR-03 | READ input

The system shall store a valid integer submitted through the GUI in the memory address specified by a READ instruction.

#### FR-04 | Input retry

The system shall preserve the pending READ instruction and prior execution state while the user corrects an invalid or out-of-range input value.

#### FR-05 | WRITE output

The system shall display each WRITE value in the GUI output area using four magnitude digits and a minus sign for negative values.

#### FR-06 | Addition

The system shall add the operand memory value to the accumulator when executing ADD.

#### FR-07 | Subtraction

The system shall subtract the operand memory value from the accumulator when executing SUBTRACT.

#### FR-08 | Multiplication

The system shall multiply the accumulator by the operand memory value when executing MULTIPLY.

#### FR-09 | Division

The system shall divide the accumulator by a nonzero operand memory value using integer division that truncates the quotient toward zero.

#### FR-10 | Unconditional branch

The system shall set the instruction counter to the operand address when executing BRANCH.

#### FR-11 | Negative branch

The system shall branch to the operand address on BRANCHNEG only when the accumulator is negative; otherwise execution shall continue sequentially.

#### FR-12 | Zero branch

The system shall branch to the operand address on BRANCHZERO only when the accumulator equals zero; otherwise execution shall continue sequentially.

#### FR-13 | Halt

The system shall stop executing instructions when it encounters HALT.

**FR-14 | Arithmetic overflow**

The system shall retain only the lowest four magnitude digits of an overflowing arithmetic result, preserving its sign, as the accumulator value used by subsequent instructions.

**FR-15 | File-error recovery**

The system shall allow the user to select another file after a file-loading error without restarting the application.

**Non-functional requirements****NFR-01 | Responsiveness**

The system shall yield to the GUI event loop after executing no more than 100 instructions in a single scheduled execution batch.

**NFR-02 | Modularity**

The system shall keep simulator and controller logic independent of Tkinter imports and GUI widget objects.

**NFR-03 | Dependencies**

The system shall run using Python 3.10 or newer with its standard library and Tcl/Tk support, without requiring third-party Python packages.